Classification of Forest Cover Type Dataset

Results and Analysis

Overview

This report presents a comparative evaluation of five supervised machine learning algorithms: Naïve Bayes, Decision Tree, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Neural Network, for the classification of forest cover types using the Forest Cover Type dataset.

The models are compared using key performance metrics, including accuracy, precision, recall, F1-score, confusion matrices, ROC curves, and Area Under the Curve (AUC). Computational time is also considered to assess the efficiency of each algorithm. These results provide a basis for identifying the relative strengths and limitations of the evaluated models.

The report further examines the suitability of the candidate algorithms in relation to the dataset's characteristics and the preprocessing techniques applied, including feature scaling, feature selection, and stratified sampling. Model generalization and the potential for overfitting are also considered when assessing their overall performance and suitability.

Based on the comparative results and these considerations, the most suitable classification algorithm is selected and its selection is justified. The report concludes with a summary of the key findings and the rationale behind the final model selection.

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1. Model Comparison

a) Accuracy

Table 1: Accuracy of each model

Model	Accuracy (%)
Naive Bayes	63.3
Decision Tree	80.6
KNN	80.8
SVM (RBF)	81.9
Neural Network (RProp MLP)	68.3

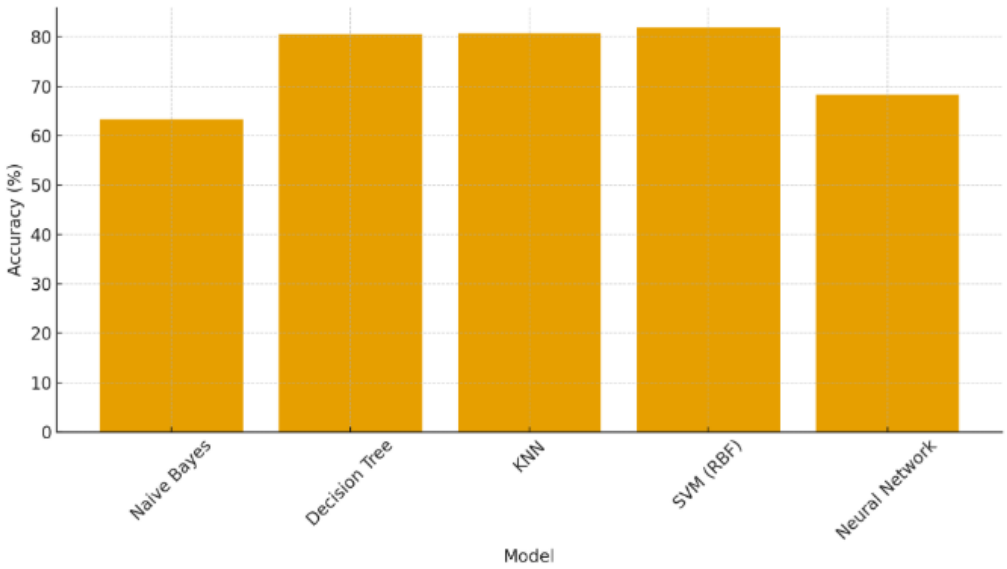


Table 2: Model Accuracy Comparison

The comparison of classifier performance based on overall accuracy shows clear differences in how effectively each algorithm predicts forest cover types.

- **SVM** (RBF) achieved the **highest accuracy** at **81.9%**, indicating that it is the most effective model for this dataset. Its ability to model non-linear decision boundaries makes it well-suited for the complex relationships among environmental features.

- **KNN** and the Decision Tree followed closely with **accuracies** of **80.8%** and **80.6%**, respectively.
  - The Decision Tree performed well due to its capability to capture non-linear splits.
  - KNN benefited from the normalized data, which ensured distance-based calculations were meaningful.

- The Neural Network (RProp MLP) achieved a moderate accuracy of 68.3%. This result is acceptable for a simple single-hidden-layer model but indicates that a deeper or more optimized architecture might improve performance.
- Naive Bayes produced the lowest accuracy at 63.3%, likely due to its strong independence assumptions, which do not align well with correlated environmental variables in the dataset.

Overall, the results show that non-linear models (such as SVM with RBF and KNN) generally perform better on this dataset.
The performance ranking from highest to lowest accuracy is:

SVM (81.9%) > KNN (80.8%) ≈ Decision Tree (80.6%) > Neural Network (68.3%) > Naive Bayes (63.3%)

This demonstrates that models capable of capturing nonlinear relationships provide superior predictive performance for forest cover classification.

b) Precision, Recall, F1-Score

Precision, recall, and F1-score values were examined for each model using the Scorer node. Since the dataset contains seven classes, KNIME does not compute a single overall precision, recall, or F1-score value, which is why the “overall” row displays missing values. Instead, evaluation was based on the per-class metrics and the confusion matrix provided for each model.

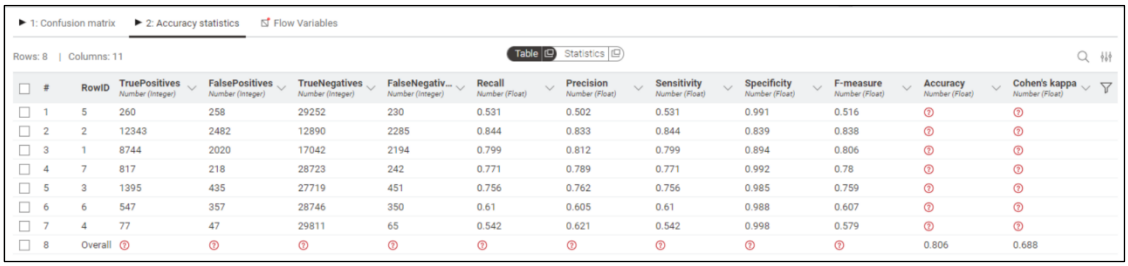


► 1: Confusion matrix ► 2: Accuracy statistics Flow Variables

Rows: 8 | Columns: 11 Table Statistics

#	RowID	TruePositives Number (Integer)	FalsePositives Number (Integer)	TrueNegatives Number (Integer)	FalseNegativ... Number (Integer)	Recall Number (Float)	Precision Number (Float)	Sensitivity Number (Float)	Specificity Number (Float)	F-measure Number (Float)	Accuracy Number (Float)	Cohen's kappa Number (Float)
1	5	57	175	29335	433	0.116	0.246	0.116	0.994	0.158		
2	2	9796	3959	11413	4832	0.67	0.712	0.67	0.742	0.69		
3	1	7519	4118	14944	3419	0.687	0.646	0.687	0.784	0.666		
4	7	189	317	28624	870	0.178	0.374	0.178	0.989	0.242		
5	3	1158	1758	26396	688	0.627	0.397	0.627	0.938	0.486		
6	6	252	559	28544	645	0.281	0.311	0.281	0.981	0.295		
7	4	14	129	29729	128	0.099	0.098	0.099	0.996	0.098		
8	Overall										0.633	0.415

Figure 1: Naive Bayes Accuracy Statistics



► 1: Confusion matrix ► 2: Accuracy statistics Flow Variables

Rows: 8 | Columns: 11 Table Statistics

#	RowID	TruePositives Number (Integer)	FalsePositives Number (Integer)	TrueNegatives Number (Integer)	FalseNegativ... Number (Integer)	Recall Number (Float)	Precision Number (Float)	Sensitivity Number (Float)	Specificity Number (Float)	F-measure Number (Float)	Accuracy Number (Float)	Cohen's kappa Number (Float)
1	5	260	258	29252	230	0.531	0.502	0.531	0.991	0.516		
2	2	12343	2482	12890	2285	0.844	0.833	0.844	0.839	0.838		
3	1	8744	2020	17042	2194	0.799	0.812	0.799	0.894	0.806		
4	7	817	218	28723	242	0.771	0.789	0.771	0.992	0.78		
5	3	1395	435	27719	451	0.756	0.762	0.756	0.985	0.759		
6	6	547	357	28746	350	0.61	0.605	0.61	0.988	0.607		
7	4	77	47	29811	65	0.542	0.621	0.542	0.998	0.579		
8	Overall										0.806	0.688

Figure 2: Decision Tree Accuracy Statistics



► 1: Confusion matrix ► 2: Accuracy statistics Flow Variables

Rows: 8 | Columns: 11 Table Statistics

#	RowID	TruePositives Number (Integer)	FalsePositives Number (Integer)	TrueNegatives Number (Integer)	FalseNegativ... Number (Integer)	Recall Number (Float)	Precision Number (Float)	Sensitivity Number (Float)	Specificity Number (Float)	F-measure Number (Float)	Accuracy Number (Float)	Cohen's kappa Number (Float)
1	5	235	125	29385	255	0.48	0.653	0.48	0.996	0.553		
2	2	12560	2640	12732	2068	0.859	0.826	0.859	0.828	0.842		
3	1	8874	2031	17031	2064	0.811	0.814	0.811	0.893	0.813		
4	7	760	161	28780	299	0.718	0.825	0.718	0.994	0.768		
5	3	1320	436	27718	526	0.715	0.752	0.715	0.985	0.733		
6	6	445	324	28779	452	0.496	0.579	0.496	0.989	0.534		
7	4	46	43	29815	96	0.324	0.517	0.324	0.999	0.398		
8	Overall										0.808	0.688

Figure 3: K-Nearest Accuracy Statistics



► 1: Confusion matrix ► 2: Accuracy statistics Flow Variables

Rows: 8 | Columns: 11 Table Statistics

#	RowID	TruePositives Number (Integer)	FalsePositives Number (Integer)	TrueNegatives Number (Integer)	FalseNegativ... Number (Integer)	Recall Number (Float)	Precision Number (Float)	Sensitivity Number (Float)	Specificity Number (Float)	F-measure Number (Float)	Accuracy Number (Float)	Cohen's kappa Number (Float)
1	5	181	46	29464	309	0.369	0.797	0.369	0.998	0.505		
2	2	13037	2996	12376	1591	0.891	0.813	0.891	0.805	0.85		
3	1	8909	1736	17326	2029	0.814	0.837	0.814	0.909	0.826		
4	7	732	94	28847	327	0.691	0.886	0.691	0.997	0.777		
5	3	1251	312	27842	595	0.678	0.8	0.678	0.989	0.734		
6	6	426	208	28895	471	0.475	0.672	0.475	0.993	0.556		
7	4	44	28	29830	98	0.31	0.611	0.31	0.999	0.411		
8	Overall										0.819	0.701

Figure 4: Support Vector Machine Accuracy Statistics



#	RowID	TruePositives Number (Integer)	FalsePositives Number (Integer)	TrueNegatives Number (Integer)	FalseNegativ... Number (Integer)	Recall Number (Float)	Precision Number (Float)	Sensitivity Number (Float)	Specificity Number (Float)	F-measure Number (Float)	Accuracy Number (Float)	Cohen's kappa Number (Float)
1	5	27	2	29508	463	0.055	0.931	0.055	1	0.104		
2	2	11508	4512	10860	3120	0.787	0.718	0.787	0.706	0.751		
3	1	7942	4261	14801	2996	0.726	0.651	0.726	0.776	0.686		
4	7	16	30	28911	1043	0.015	0.348	0.015	0.999	0.029		
5	3	956	644	27510	890	0.518	0.598	0.518	0.977	0.555		
6	6	53	49	29054	844	0.059	0.52	0.059	0.998	0.106		
7	4	0	0	29858	142	0		0	1			
8	Overall										0.683	0.461

Figure 5: Neural Network Accuracy Statistics

Across the models, the higher-performing classifiers (SVM, KNN, and the Decision Tree) exhibited consistently higher precision and recall for the majority of classes, especially those with more training samples. These models generally achieved higher F1-scores for the major classes, demonstrating a balanced ability to correctly identify and retrieve class instances.

Naive Bayes showed the largest drop in precision and recall, particularly for classes with overlapping feature distributions or correlated inputs, leading to lower F1-scores and more frequent misclassifications. The Neural Network achieved moderate precision and recall but showed instability in minority classes due to its simple architecture.

Overall, the models with higher accuracy (SVM, KNN, and Decision Tree) also showed stronger per-class precision, recall, and F1 performance, confirming that these classifiers not only predict correctly overall but also maintain balanced performance across multiple forest cover types.

c) Confusion Matrix

The confusion matrices of the five models reveal clear differences in how effectively each classifier distinguishes between forest cover types.

Naive Bayes shows relatively high numbers of false positives (FP) across most classes. This indicates that the model frequently misclassifies instances as belonging to certain classes when they do not. It does, however, maintain more true negatives (TN) than false negatives (FN) in all classes, meaning it is better at correctly identifying non-membership than detecting the correct class.

In contrast, Decision Tree, KNN, and SVM demonstrate much stronger class discrimination. All three models consistently show more true positives (TP) than false positives (FP) and more true negatives (TN) than false negatives (FN) across every

class. This pattern indicates that these models are more effective at both correctly identifying class members and avoiding incorrect classifications.

The Neural Network (RProp MLP) shows higher TP than FP for most classes, although not as consistently as SVM, KNN, or the Decision Tree. Like the other models, it maintains TN > FN for all classes, indicating reliable performance in identifying non-membership even if its class-specific accuracy varies.

Overall, the confusion matrix analysis confirms that SVM, KNN, and the Decision Tree provide the strongest per-class performance, while Naive Bayes struggles due to its independence assumptions and Neural Network provides moderate but less stable class discrimination.

d) ROC Curve and AUC

ROC and AUC metrics were not available because KNIME does not compute these measures for multi-class (7-class) classification problems. As a result, the models were evaluated using accuracy, precision, recall, F1-score, and confusion matrices instead.

e) Computational Time

Table 3: Computational time of each model

Model	Computational Time (Qualitative)
Naive Bayes	Very Fast
Decision Tree	Fast
KNN	Slow Prediction
SVM (RBF)	Slowest Training
Neural Network	Moderate

Computational time was evaluated qualitatively based on how long each model required to train and generate predictions within KNIME. Naive Bayes was the fastest algorithm because it relies on simple probability calculations and does not require iterative optimization. The Decision Tree model also trained quickly due to its straightforward splitting process.

KNN has no training phase, but its prediction step is slower because the algorithm must compute distances to all training instances. SVM with the RBF kernel had the longest training time, as it needs to optimize a non-linear decision boundary in a high-dimensional feature space. The Neural Network (RProp MLP) required moderate training time due to multiple weight-updating iterations, but still ran efficiently for the chosen network size.

Overall, Naive Bayes and Decision Tree were the fastest models, while SVM and KNN required more computation, and the Neural Network fell in the middle.

2. Best Algorithm Selection

Based on the comparative evaluation, the Support Vector Machine (SVM) with the RBF kernel achieved the highest accuracy (81.9%) and demonstrated the strongest overall performance. Several factors contribute to SVM's superior results on this dataset.

a) Suitability for Non-Linear Relationships

The forest cover type dataset contains complex, non-linear interactions between features such as elevation, slope, soil type, horizontal distances, and hillshade values.
The RBF kernel is specifically designed to model non-linear decision boundaries, allowing SVM to separate classes that cannot be distinguished using simple linear splits.
This gives it an advantage over algorithms like Naive Bayes and linear classifiers.

b) Benefit from Feature Scaling (Normalization)

SVM is highly sensitive to feature scales, and your dataset was normalized during preprocessing.
This significantly improves the performance of SVM because:

- All features contribute equally
- The margin calculations become more stable
- The kernel function operates effectively

The normalization step directly helped SVM achieve better results.

c) Reduced Dimensionality from Feature Selection

You applied correlation-based feature reduction, which removed redundant and highly correlated columns. SVM benefits strongly from:

- reduced noise
- fewer irrelevant features
- tighter, cleaner decision boundaries

Reducing dimensionality made the SVM model more robust and less prone to overfitting.

d) Balanced Class Representation (Stratified Sampling)

Using stratified sampling ensured each class was represented proportionally in training and testing.

For SVM, which relies on support vectors near class boundaries, balanced representation is essential for stable performance.

e) Strong Generalization and Low Overfitting

Unlike Decision Trees, which can easily overfit, SVM is designed to maximize the margin between classes, resulting in better generalization. This explains why:

- Decision Tree had similar accuracy (~80.6%) but slightly lower generalization
- KNN performed well (80.8%) but is more sensitive to noise and redundant features
- SVM outperformed both

Overall, the SVM model performed the best because it effectively captured the non-linear structure of the dataset, benefited strongly from preprocessing steps such as normalization and correlation-based feature selection, and maintained good generalization without overfitting. Its ability to create flexible decision boundaries using the RBF kernel allowed it to outperform the other classifiers, particularly on a heterogeneous multi-class dataset like forest cover type.

3. Conclusion

This project applied five different machine learning algorithms—Naive Bayes, Decision Tree, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and a Neural Network (RProp MLP)—to predict forest cover types using a preprocessed and feature-engineered dataset. Several steps were taken to enhance model performance, including normalization, correlation-based feature selection, removal of irrelevant

attributes, and stratified sampling to ensure balanced class distribution. These preprocessing steps contributed significantly to the stability, accuracy, and generalization of the models.

Among the five classifiers, the SVM model with an RBF kernel delivered the best overall performance with an accuracy of 81.9%, demonstrating strong capability in modeling the non-linear structure of the dataset. KNN and the Decision Tree also performed competitively, achieving accuracies above 80%, showing that distance-based and tree-based methods are well-suited for environmental datasets with mixed numerical characteristics. The Neural Network performed moderately (68.3%), indicating that a more complex architecture or hyperparameter optimization may be required for improved results. Naive Bayes produced the lowest accuracy (63.3%), reflecting the limitations of its independence assumptions when applied to correlated environmental features.

Evaluation using precision, recall, F1-score, and confusion matrices further confirmed these trends, with SVM, KNN, and Decision Tree showing stronger class-level performance and more balanced predictions across the seven cover types. ROC curves and AUC were not generated due to the multi-class nature of the dataset. Computational time analysis also showed clear distinctions: Naive Bayes and Decision Tree were fastest, SVM and KNN required more time, and the Neural Network showed moderate training time.

Overall, this study concludes that non-linear models, particularly SVM with an RBF kernel, are highly effective for forest cover type prediction when combined with appropriate preprocessing. Future improvements could include hyperparameter tuning, cross-validation, more complex neural network architectures, or ensemble methods to further boost model performance.